

Lab 02 - Processamento de Bases de Dados em Lote

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```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

Atribuindo as funções getstats e computestats

```
getStats <- function(input, pos) {
  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) %>%
    filter(!is.na(ARRIVAL_DELAY)) %>%
    group_by(MONTH, DAY, AIRLINE) %>%
    summarise(
      total_voos = n(),
      voos_atrasados = sum(ARRIVAL_DELAY > 10)
    )
}

computeStats <- function(stats) {
  stats %>%
    group_by(MONTH, DAY, AIRLINE) %>%
    summarise(
      total_voos = sum(total_voos),
      voos_atrasados = sum(voos_atrasados)
    ) %>%
    mutate(
      percentual = round((voos_atrasados / total_voos) * 100, 2)
    )
}
```

Processando em lotes

```
Estats_por_chunk <- read_csv_chunked("flights.csv.zip",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 100000,
```

```
col_types = cols_only(
  MONTH = col_integer(),
  DAY = col_integer(),
  AIRLINE = col_character(),
  ARRIVAL_DELAY = col_integer())
)
```

`summarise()` has grouped output by 'MONTH', 'DAY'. You can override using the
`.groups` argument.

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Encontrando o resultado final

```
result_final <- computeStats(Estats_por_chunk)

`summarise()` has grouped output by 'MONTH', 'DAY'. You can override
using the
`.groups` argument.

head(result_final)

# A tibble: 6 × 6
# Groups:   MONTH, DAY [2]
  MONTH    DAY AIRLINE total_voos voos_atrasados percentual
  <int> <int> <chr>      <int>          <int>         <dbl>
1     1     1    AA        1379            514        37.3
2     1     1    DL        1558            132         8.47
```

3	1	1	UA	1269	367	28.9
4	1	1	US	1028	201	19.6
5	1	2	AA	1508	649	43.0
6	1	2	DL	2261	406	18.0

Analisando o total por companhia aérea

```
por_companhia <- result_final%>%
  group_by(AIRLINE)%>%
  summarise(total_voos_gerais= sum(total_voos),
            voos_atrasados_gerais = sum(voos_atrasados)) %>%
  mutate(percentual_de_atraso =
round((voos_atrasados_gerais/total_voos_gerais)*100, 2))%>%
  arrange(percentual_de_atraso)
```

```
head(por_companhia)
```

```
# A tibble: 4 × 4
  AIRLINE total_voos_gerais voos_atrasados_gerais percentual_de_atraso
  <chr>          <int>          <int>          <dbl>
1 DL             870275          141889          16.3
2 AA             712935          154114          21.6
3 US             194223           44307          22.8
4 UA             507762          121088          23.8
```

momento de compilação

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 15:54:17 -03"
```